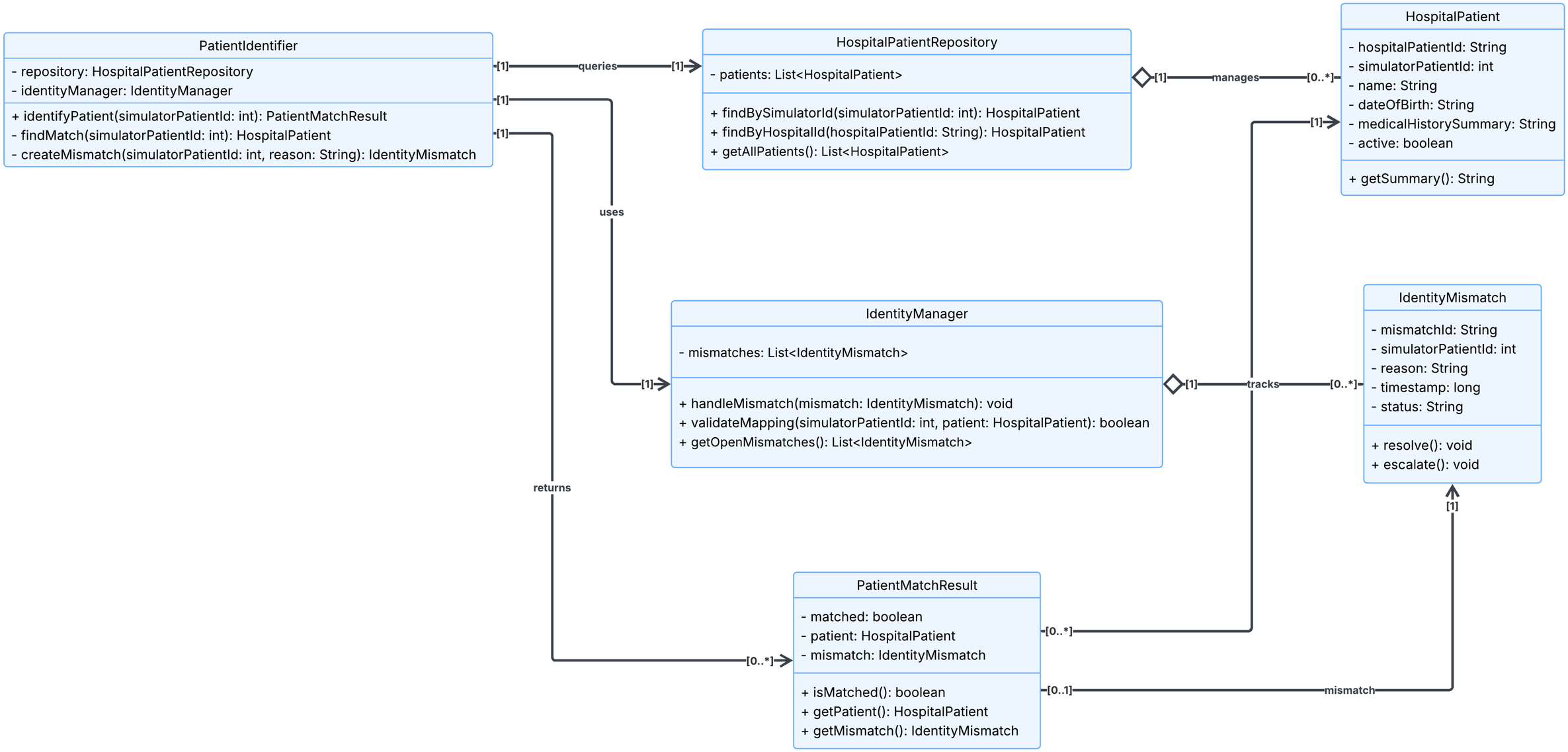
CHMS Patient Identification System - Class Diagram



This diagram models how the CHMS links simulator-generated patient IDs to real hospital patient records. The design separates patient retrieval, identity validation, and anomaly handling so that the system remains reliable, traceable, and easier to maintain.

PatientIdentifier is the central matching component. It receives a simulator patient ID, queries HospitalPatientRepository, and returns a PatientMatchResult. This keeps the matching workflow in one place. HospitalPatientRepository stores and retrieves HospitalPatient records, which separates record access from identity validation logic. HospitalPatient contains the real patient details needed by the hospital system, such as name, date of birth, medical history summary, and active status.

IdentityManager is responsible for validating mappings and handling edge cases. This avoids overloading the identifier with mismatch management. IdentityMismatch represents problems such as missing patients, duplicate mappings, or inactive records, which makes anomalies explicit and traceable. PatientMatchResult wraps the identification outcome by storing either a matched patient or a mismatch result. This is cleaner than returning raw null values and supports future extension.

Overall, the subsystem emphasizes clear matching logic, controlled access to patient identity data, and structured handling of anomalies, which fits the needs of a realistic hospital monitoring system.